

AushadhiSetu

Last-Mile PHC Stock Guardian (औषधि सेतु)

Track 3 Smart Health | Build with AI Code for Communities 2.0

Demo: <https://aushadhisetu-blond.vercel.app/>

The Problem

- Rural pharmacists are overwhelmed by manual paper registers.
- DVDMS digital entry is delayed, causing critical systemic blind spots.
- Missed tracking leads directly to stockouts of essential medicines.

**"CAG Rajasthan 2022:
27% of PHCs < 50% essential
drugs."**

**"Maharashtra IT Audit:
Demand module completely
unused."**

Personas

- **Pharmacist:** Overwhelmed by 100+ manual daily entries per register.
- **MOIC:** Approves indent requests blindly without data backing.
- **District Officer:** Lacks real-time visibility across PHCs.



Pharmacist "I have no time to type."



MOIC "Are we over-ordering?"



State "Where are the shortages?"

Existing System vs. Our Gap Fill

- We are **NOT** replacing the state DVDMS system.
- We digitize the "first mile" (paper register) via AI OCR.
- Accurate data flows to DVDMS faster, preventing stockouts.

Feature	State DVDMS	AushadhiSetu
Level	State / District	Local PHC
Input	Manual Keyboard	AI Photo / Voice
Speed	Delayed (Days)	Real-Time

Solution Flow

- Photo or Voice input captures register data effortlessly.
- Gemini 2.5 AI accurately extracts data to strict JSON.
- Local JS Forecast calculates risk and drafts MOIC Indent.

 Paper Register /  Voice



 Gemini 2.5 Flash AI



 Local JS Forecast



 Human Approved Indent

The AI Engine (Vision & Voice)

- Gemini 2.5 Flash via AI Studio processes messy registers.
- Outputs strict JSON schema with validation checksums.
- Understands spoken Hindi for quick stock updates.

```
{
  "drugCode": "DRG001",
  "name": "Paracetamol 500mg",
  "opening": 500,
  "issued": 50,
  "closing": 450,
  "_checksum": true,
  "confidence": {
    "closing": 0.96
  }
}
```

Risk Board (Live Demo Proof)

- Local JS forecasting dynamically categorizes 60 drugs.
- Real-time alerts for stockouts and approaching expiry.
- **Verified Demo State:** 7 RED / 15 AMBER / 38 GREEN.

● Paracetamol 500mg

0 - 9 Days Left | Stockout Imminent

● Amoxicillin 250mg

1 - 12 Days Left | Critical Need

● ORS Sachet

Overstock | Batch expires in 60 days

● Iron Folic Acid

Safe | > 30 Days Stock

Data Architecture & Disclosures

- 100% Vercel free tier (Next.js PWA + API Routes).
- Local seed.json prevents heavy DB costs.
- Completely transparent AI assistance & guardrails.

Hackathon Architecture Disclosures

- **Synthetic Data:** Uses generated Demo Data.
- **Local Compute:** JS Forecast used (Not BQML).
- **No State API:** No live DVDMS integration.
- **Ethics:** Zero PII (Patient Info) processed.
- **Safety:** MOIC manual approval mandatory.

Pilot, Scale, & Cost

- Ready for an immediate 30-day pilot deployment.
- Edge-compute architecture keeps infrastructure minimal.
- Massive state savings by preventing emergency bulk purchases.

Vercel Free Tier
Rs. 0 / PHC

Safeguards & Ethics

- **Privacy First:** We track drug volumes, never patient names.
- **Human-in-the-Loop:** AI only assists, never decides.
- **Control:** Auto-ordering is strictly hardcoded OFF.

Final Indent Approval



I (MOIC) approve this indent manually.

Team, Stack & Open Source

- Built with **Next.js**, **Vercel**, and **Gemini 2.5 Flash**.
- Fully transparent and modifiable architecture.
- Open-source license for community adoption.

Source Code Available

github.com/bc2ykys/aushadhisetu

Licensed under Apache-2.0

Try It Now

Scan to open the Vercel PWA Demo



<https://aushadhisetu-blond.vercel.app/>

Thank you, Google Cloud & Hack2Skill!